EXPT NO: 3	ASSIGNMENT PROBLEM
DATE:	

AIM:

To understand and solve the assignment problem using python ,assignment team of workers.

PROCEDURE:

1. Input Data: Define the cost matrix where each element represents the cost of assigning a particular task to a specific worker.
2. Algorithm: Use an optimization algorithm like the Hungarian algorithm (or the Munkres algorithm) to find the optimal assignment that minimizes the total cost.
3. Output: Display the optimal assignment along with the minimum total cost.

CODE

PROGRAM 1:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Cost matrix
cost_matrix = np.array([
    [15, 10, 9],
    [9, 15, 10],
    [10, 12, 8]
])

print("Cost Matrix:")
print(cost_matrix)

# Solve assignment problem
row_ind, col_ind = linear_sum_assignment(cost_matrix)

total_cost = 0
```

```
print("\nOptimal Assignment:")


for r, c in zip(row_ind, col_ind):
    print(f"Person {r+1} -> Job {c+1} | Cost = {cost_matrix[r][c]}")
    total_cost += cost_matrix[r][c]

print("\nMinimum Total Cost =", total_cost)
```

OUTPUT:

```
Cost Matrix:
[[15 10  9]
 [ 9 15 10]
 [10 12  8]]

Optimal Assignment:
Person 1 -> Job 2 | Cost = 10
Person 2 -> Job 1 | Cost = 9
Person 3 -> Job 3 | Cost = 8

Minimum Total Cost = 27
```

PROBLEM 2:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Cost matrix
cost_matrix = np.array([
    [11, 7, 10],
    [9, 8, 7],
    [6, 10, 12]
])

print("Cost Matrix:")
print(cost_matrix)

# Solve assignment problem
row_ind, col_ind = linear_sum_assignment(cost_matrix)

total_cost = 0
print("\nOptimal Assignment:")

for r, c in zip(row_ind, col_ind):
```

```
    print(f"Person {r+1} -> Job {c+1} | Cost = {cost_matrix[r][c]}")
    total_cost += cost_matrix[r][c]

print("\nMinimum Total Cost =", total_cost)
```

OUTPUT:

Cost Matrix:
[[11 7 10]
[9 8 7]
[6 10 12]]

Optimal Assignment:
Person 1 -> Job 2 | Cost = 7
Person 2 -> Job 3 | Cost = 7
Person 3 -> Job 1 | Cost = 6

Minimum Total Cost = 20

RESULT:

The Assignment Problem has been completed successfully.